

Machine learning in packet routing process using Quagga/Zebra routing SW suite

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Agenda

1. Link-state routing protocols
2. Where is the room for improvement?
3. Current LS routing protocols implementation
4. How to use ML in current OSFP/IS-IS networks?
5. Results in simulated environments
6. Potential practical applications and limitations

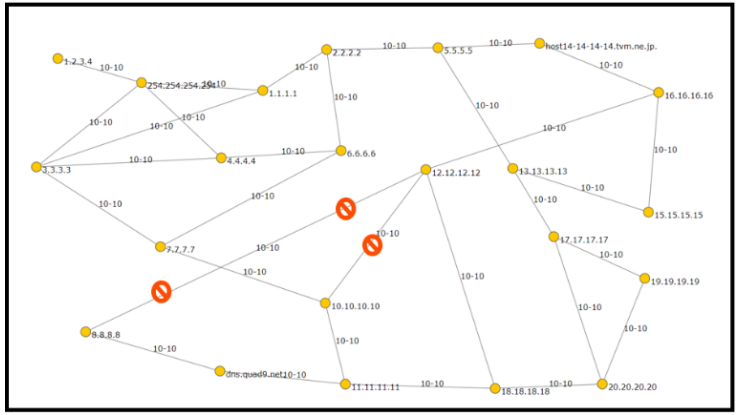
Link-state routing protocols

1. Link-state routing protocols are based on Dijkstra's algorithm (Shortest Path First), which is a deterministic and 100% accurate algorithm for network routes calculation.
2. Most popular: OSPF, IS-IS
3. Question:

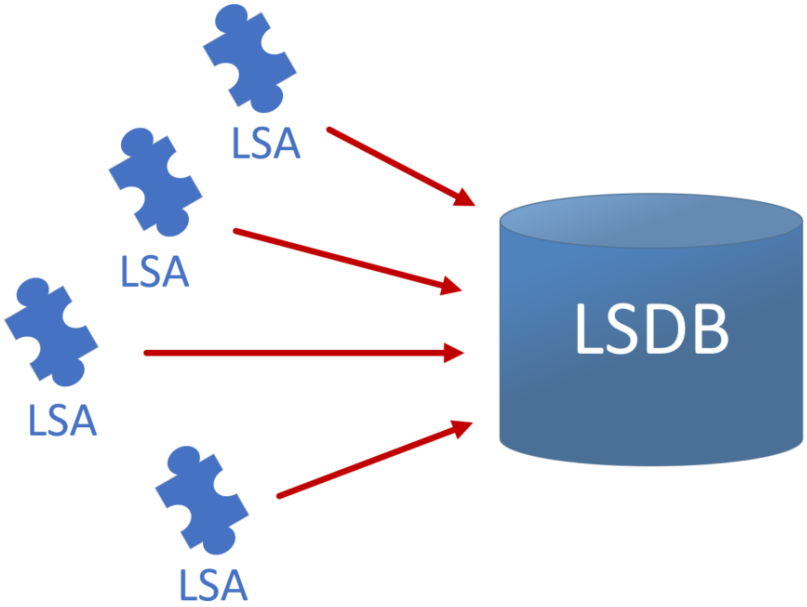
   Where is the room for improvement in those networks?

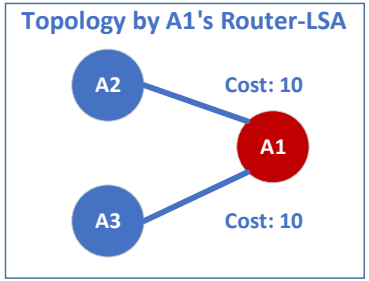
Room for impromevent in SFP protocols

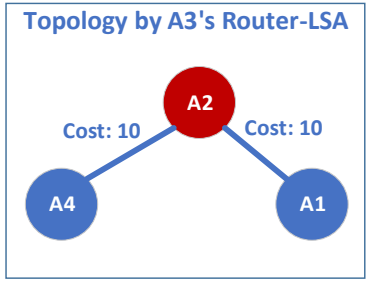
1. Modern networks are getting bigger, more virtualized and more dynamic.
2. Tons of VM are being added or removed every minute.
3. **Despite dynamic environment we expect stable and 100% efficient network.**
4. However… in case of multiple links failures the time of adaptation in complex scenario may takes even seconds.
5. The idea is to REDUCE the time of the network recovery (self-healing time) in case of multiple link failures.

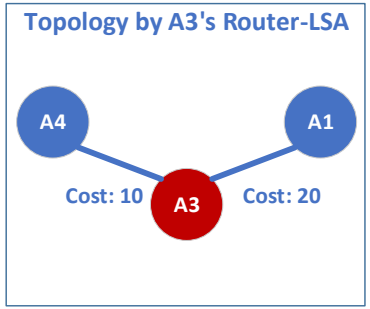


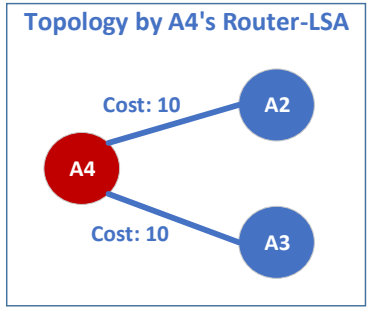
Link State DataBase (LSDB)

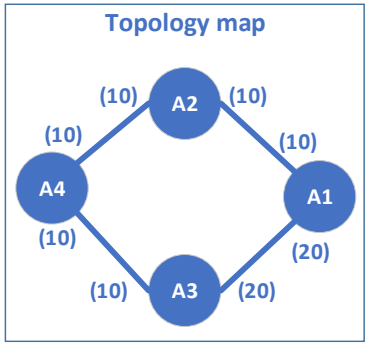








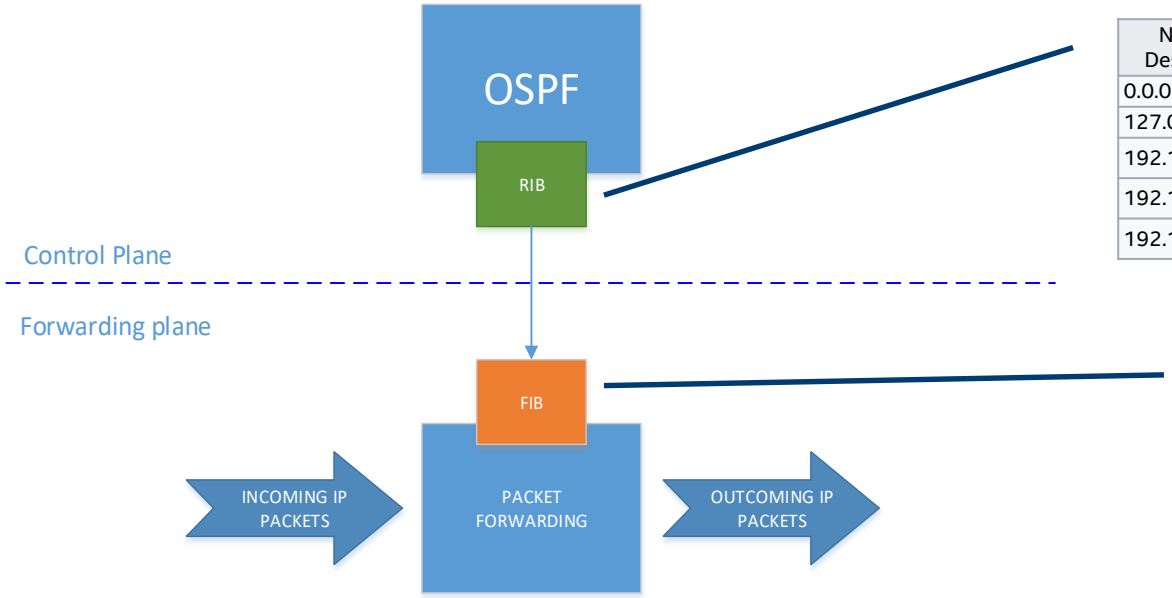




LSDB

Originator	Neighbor	Cost
A1	A2	10
A1	A3	10
A2	A4	10
A2	A1	10
A3	A4	10
A3	A1	20
A4	A2	10
A4	A3	10

Routing and Forwarding tables



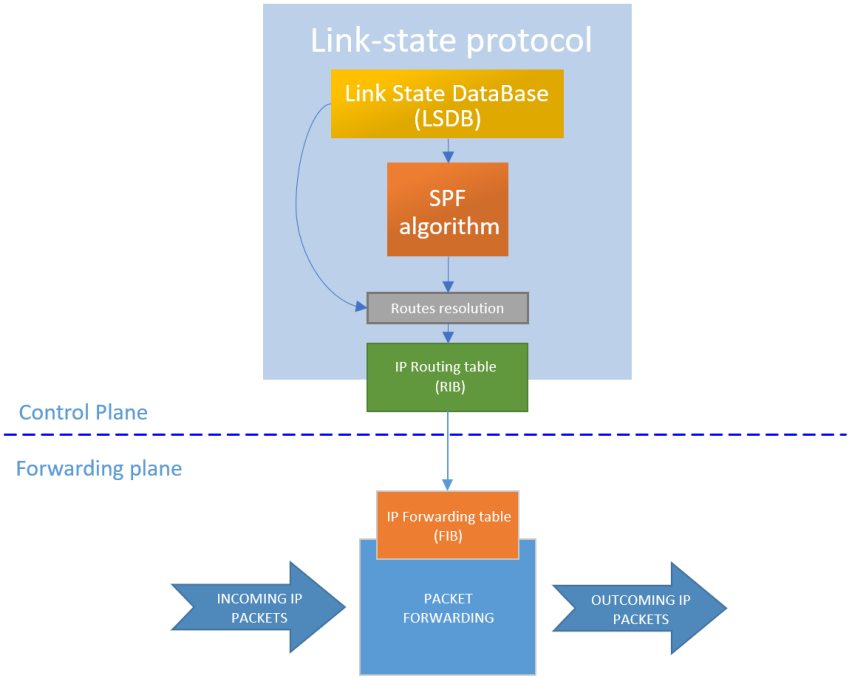
IP Routing table (RIB):

Network Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	192.168.0.1	192.168.0.100	10
127.0.0.0	255.0.0.0	127.0.0.1	127.0.0.1	1
192.168.0.0	255.255.255.0	192.168.0.100	192.168.0.100	10
192.168.0.100	255.255.255.255	127.0.0.1	127.0.0.1	10
192.168.0.1	255.255.255.255	192.168.0.100	192.168.0.100	10

IP Forwarding table (FIB):

Prefix	Next Hop	Interface
0.0.0.0/0	192.168.0.1	Eth0
127.0.0.0/32	Receive	n/a
192.168.0.0/24	192.168.0.10	Eth0
192.168.0.100/24	Receive	n/a
192.168.0.1/24	192.168.0.10	Eth0

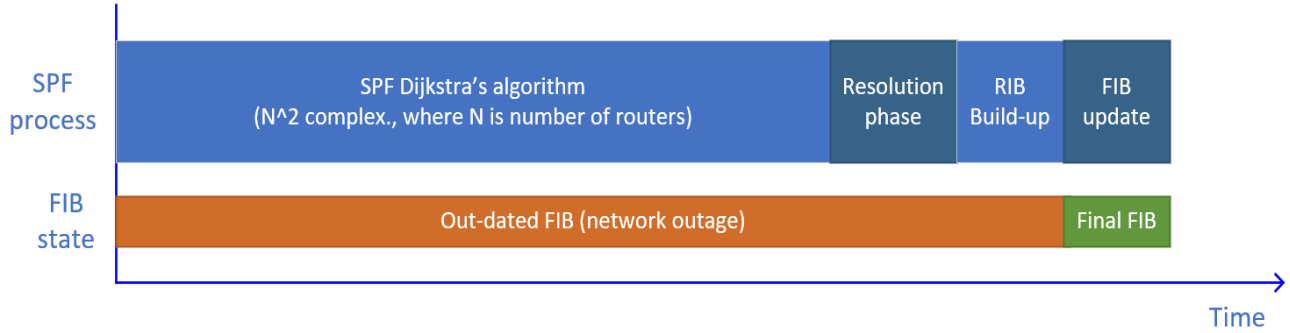
Current LS routing protocols implementation



How route calculations are done in current implementations (FRR, Quagga)?

1. SPF calculations (each router builds a tree to reach the other routers)
2. Route resolutions (dependent routes are then resolved, choosing what router to reach in order to route to that destination)
3. Routing table build-up (RIB)

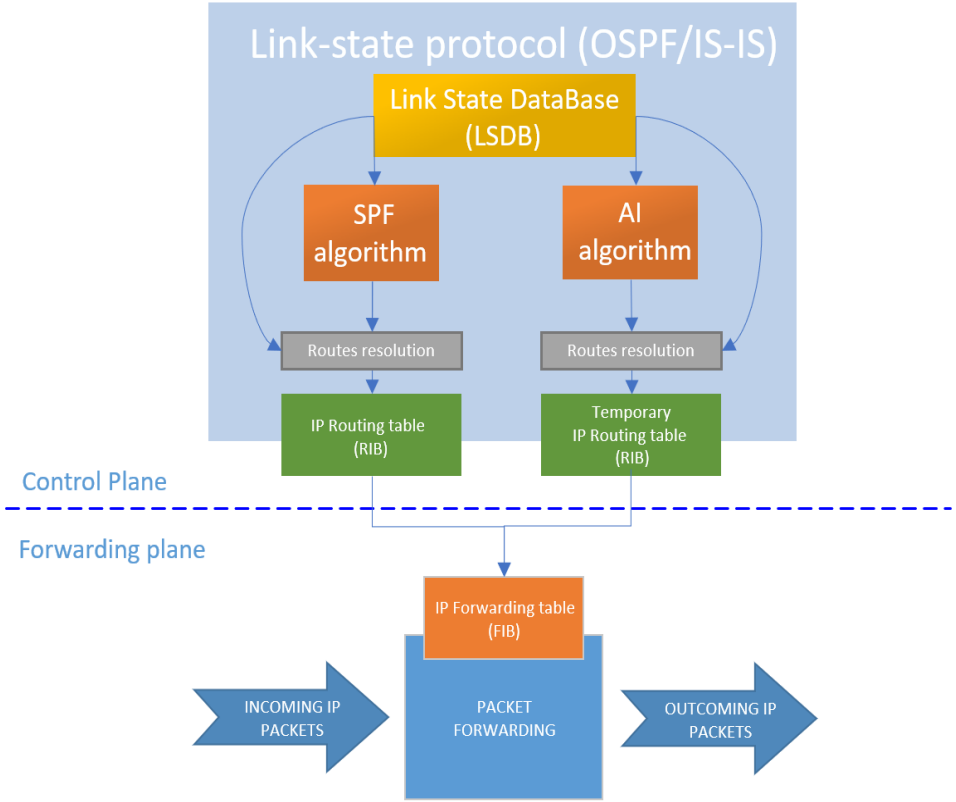
Disadvantages of current implementations



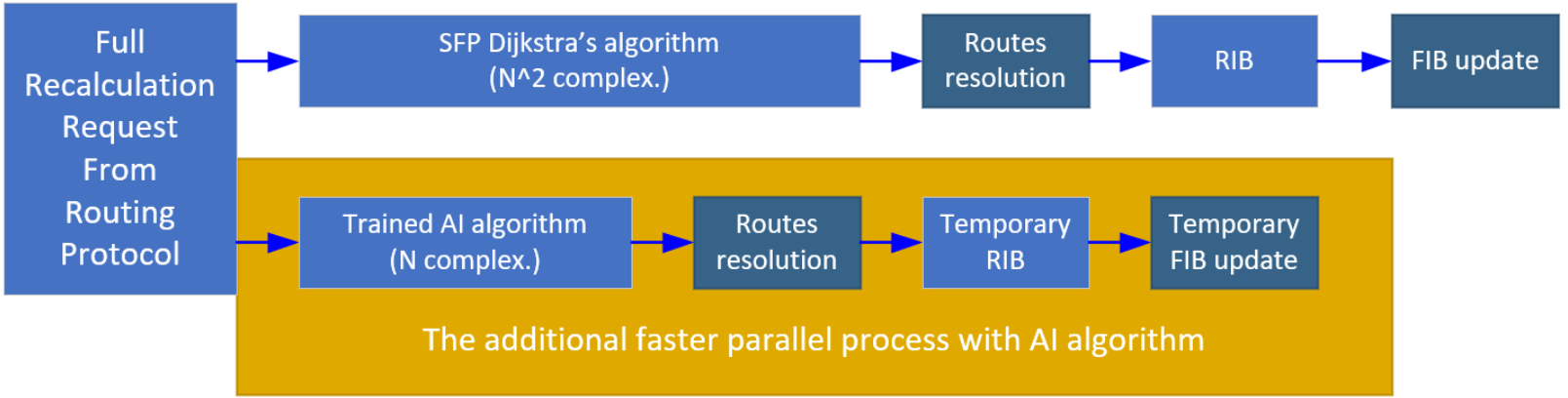
- In current SPF implementations it can be run in Full, Partial or Incremental mode.
- Big topology changes requires full SPF recalculation.
- During the full SPF recalcution we can observe network outage (in orange).

How to use ML in current OSFP/IS-IS networks?

- We want to use ML to enhance the RIB recalculation step.
- The plan is to add parallel ”AI algorithm” along with ”Temporary RIB”.
- AI algorithm is trained to output the next hop from the local router to reach each other router in the network.

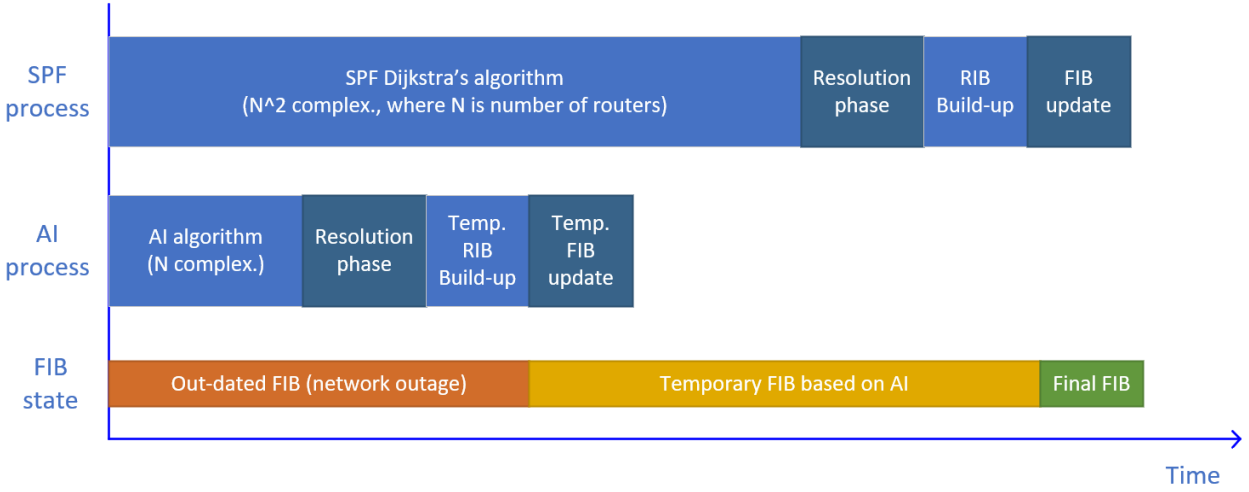


Potential usage of ML in current implementation



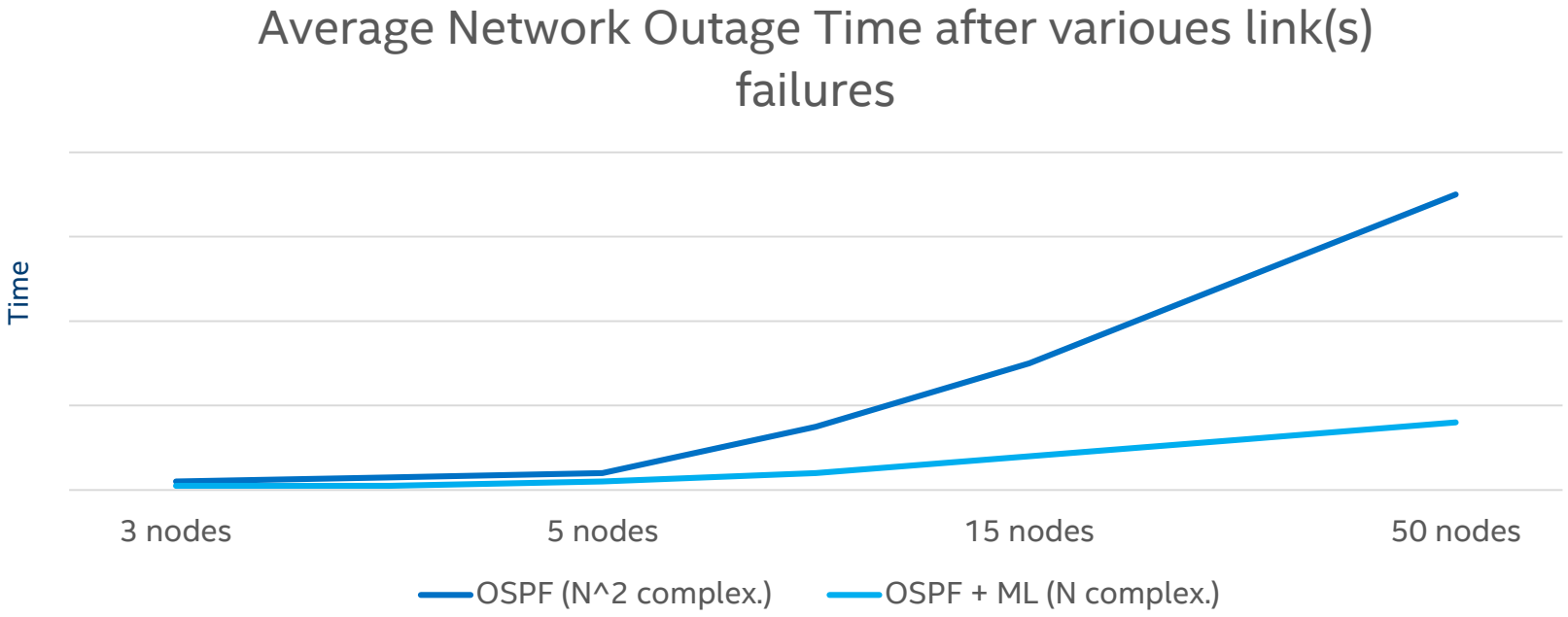
- Using ML we want to generate the temporary FIB in case of network problems that need the full SPF recalculation.
- The AI algorithm, for example a Neural Network, is trained based on previously calculated SPF routes, or on a set of datasets generated in a simulated environment.

Potential usage of ML in current implementation



- The overall goal is to reduce the time needed to have a **temporary FIB** that is, with high probability, good enough for the topological state the network is in after the topological change.
- **Temporary FIB** is used until the full SPF recalculation has come to its end.

Tests in simulated environments



Potential practical applications and limitations.

Potential utilizations and application:

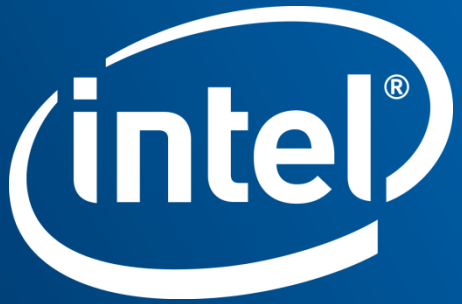
- Industrial size data centers
- Even one ”intelligent” router node can decrease average overall network self-healing time.
- Can be implemented in FRR, Quagga or any other SW routing suite.

Limitations:

- Will not resolve congestion problems.



THANK YOU !!!


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